T.C.

GEBZE TECHNICAL UNIVERSITY

PHYSICS DEPARTMENT

PHYSICS LABORATORY I

EXPERIMENT REPORT

THE NAME OF THE EXPERIMENT

Determining the restoring torque of the torsion axle

GEBZE TEKNİK ÜNİVERSİTESİ

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Experimental procedure for determining the restoring torque of the torsion axle

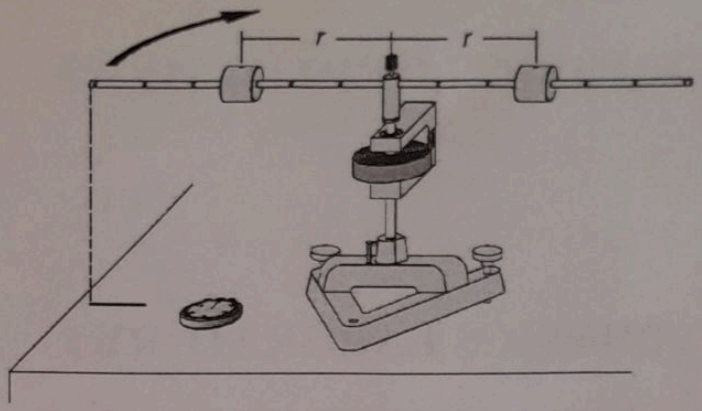
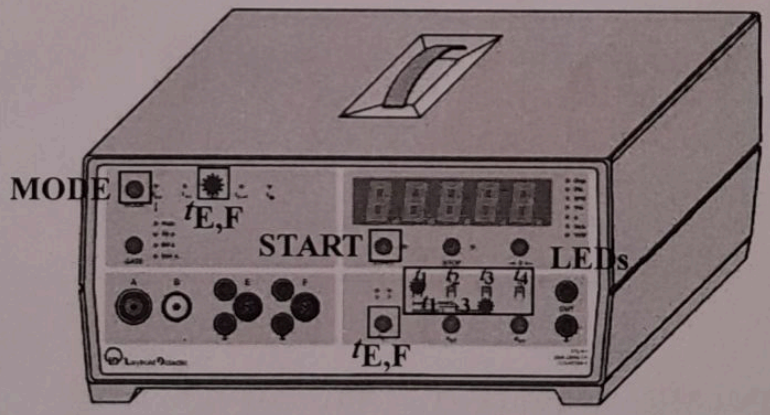


Figure 6.2: Experimental setup for determining the period of oscillation.

The experimental setup is illustrated in Fig. 6.2.

1. Fix the middle of the transverse rod to the torsion axle arrange the weights symmetrically at a distance of 10 cm from the torsion axle.
2. Locate the transverse rod so that it is in the photo gate in its equilibrium (rest) position.
3. Set the operation *MODE* on the digital counter to the symbol $t_{E,F}$ for measuring the period.
4. Rotate the transverse rod to the right by < 180°, press the start button of the digital counter and release the rod. Wait until all the display of storage LED's light up.



5. Read the period T by pressing pushbutton $t_{E,F}$ and select of the time $t_{1\to3}$ stored in the operation mode $t_{E,F}$ to be displayed as the measured T value. *In the storage display, the LED corresponding to the selected time shines more brightly than the others.*
6. Repeat the measurement three times, alternately deflecting the rod to the left and to the right.
7. One after another increase the distance to 15, 20, 25 and 30 cm, each time repeating the measurement.
8. Finally, remove the weights, and repeat the measurement for the transverse rod. Record the measured periods to the table 6.1.
9. Measure the mass of the weights m 236 gr gr.

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Calculate mean value of measured period of oscillation T and T^2 and fill in the Table 6.1.

Table 6.1: Period of oscillation T for a transverse rod with weights attached at a distance r from the axis of rotation.

r (m)	r^2 (m^2)	T_1 (s)	T_2 (s)	T_3 (s)	T_4(s)	T (s)	T^2(s^2)
0.10	0,01	4,259	4,258	4,261	4,261	4,260	18,148
0.15	0,0225	5,510	5,504	5,504	5,503	5,505	30,305
0.20	0,04	6,890	6,884	6,881	6,885	6,885	47,403
0.25	0,0625	8,333	8,332	8,335	8,341	8,335	69,472
0.30	0,09	9,784	9,780	9,781	9,780	9,781	95,668
Without weights	0	2,892	2,893	2,894	2,891	2,892	8,364

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